# Nathan Wallheimer

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Rehovot, Israel



## Education

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| 2022–present | PhD in Computer Science (in progress), Weizmann Institute of Science.<br>Advisor: Prof. Amir Abboud.<br>Thesis: *Structure versus Pseudorandomness in Fine-Grained Complexity.* |
| 2018–2021 | MSc in Computer Science, University of Haifa.<br>Advisor: Prof. Oren Weimann. |
| 2014–2018 | BSc in Computer Science, University of Haifa. |

## Professional Experience and Activities

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| July 2024 | Teaching Assistant, DIMACS, Rutgers University. |
| 2017–2021 | Teaching Assistant, Department of Computer Science, University of Haifa. |
| 2016–2018 | Private tutor, academic support program for BSc students experiencing academic difficulties, University of Haifa. |

**Conference presentations:** ISAAC 2022, ITCS 2023, HALG 2023, ESA 2024, SODA 2025, ITCS 2026, and HALG 2026.

**Reviewer for:**

- IEEE Symposium on Foundations of Computer Science (FOCS).
- ACM-SIAM Symposium on Discrete Algorithms (SODA).
- International Colloquium on Automata, Languages, and Programming (ICALP).
- Innovations in Theoretical Computer Science Conference (ITCS).
- European Symposium on Algorithms (ESA).
- Scandinavian Symposium on Algorithm Theory (SWAT).
- International Symposium on Mathematical Foundations of Computer Science (MFCS).
- Annual Symposium on Combinatorial Pattern Matching (CPM).

## Teaching and Academic Service

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| 2022–present | Algorithms Seminar Organizer, Weizmann Institute of Science.<br>Select speakers, coordinate logistics, and host guests. |
| July 2024 | DIMACS Tutorial on Fine-Grained Complexity, Rutgers University. |
| 2018–2021 | Data Structures, University of Haifa. |
| 2017–2021 | Advanced Data Structures and Probability, University of Haifa. |

## Honours and Awards

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| 2021 | Dean's List, University of Haifa (MSc in Computer Science). |
| 2019–2020 | Excellence scholarship, University of Haifa. |

**Languages:** Hebrew, English, French.

## Publications

1. Amir Abboud, Shyan Akmal, Ron Safier, and Nathan Wallheimer.
   **Lower Bounds for Structured OMv, APSP, and Triangle Detection.**
   *Innovations in Theoretical Computer Science Conference (ITCS 2027).* To appear.

2. Amir Abboud, Ron Safier, and Nathan Wallheimer.
   **Equivalent Dichotomies for Triangle Detection in Subgraph, Induced, and Colored H-Free Graphs.**
   In *34th Annual European Symposium on Algorithms (ESA 2026)*, LIPIcs 388, pp. 64:1–64:22. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2026. doi:10.4230/LIPIcs.ESA.2026.64.

3. Keren Censor-Hillel, Tomer Even, Virginia Vassilevska Williams, and Nathan Wallheimer.
   **Witness-Sensitive Detection of Induced Diamonds.**
   In *53rd International Colloquium on Automata, Languages, and Programming (ICALP 2026)*, LIPIcs 374, pp. 52:1–52:22. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2026. doi:10.4230/LIPIcs.ICALP.2026.52.

4. Amir Abboud, Ron Safier, and Nathan Wallheimer.
   **Triangle Detection in H-Free Graphs.**
   In *17th Innovations in Theoretical Computer Science Conference (ITCS 2026)*, LIPIcs 362, pp. 1:1–1:19. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2026. doi:10.4230/LIPIcs.ITCS.2026.1.

5. Amir Abboud, Nick Fischer, Ron Safier, and Nathan Wallheimer.
   **Recognizing Sumsets is NP-Complete.**
   In *Proceedings of the 2025 Annual ACM-SIAM Symposium on Discrete Algorithms (SODA)*, pp. 4484–4506. SIAM, 2025. doi:10.1137/1.9781611978322.153.

6. Amir Abboud and Nathan Wallheimer.
   **Worst-Case to Expander-Case Reductions: Derandomized and Generalized.**
   In *32nd Annual European Symposium on Algorithms (ESA 2024)*, LIPIcs 308, pp. 4:1–4:18. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2024. doi:10.4230/LIPIcs.ESA.2024.4.

7. Amir Abboud and Nathan Wallheimer.
   **Worst-Case to Expander-Case Reductions.**
   In *14th Innovations in Theoretical Computer Science Conference (ITCS 2023)*, LIPIcs 251, pp. 1:1–1:23. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2023. doi:10.4230/LIPIcs.ITCS.2023.1.

8. Shay Mozes, Nathan Wallheimer, and Oren Weimann.
   **Improved Compression of the Okamura-Seymour Metric.**
   In *33rd International Symposium on Algorithms and Computation (ISAAC 2022)*, LIPIcs 248, pp. 27:1–27:19. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2022. doi:10.4230/LIPIcs.ISAAC.2022.27.